South Australia's Blue Bio Industry

Investigate South Australia's unique marine biota with our world–leading research capability

South Australia is home to well-established fisheries and aquaculture industries, with a reputation for producing safe, sustainable, high quality and high value products in both land and marine based environments.

With 80% of the state's marine life occurring nowhere else, leading research and development, and proactive regulation, the opportunity exists to explore and develop new products from our bioresources.

South Australia is a great place to live, work and play. The capital, Adelaide, is consistently ranked as one of the most liveable cities in the world.



Nearly 80% of South Australia's marine biota is unique to our region.

Southern Australia has over 1,400 species of seaweed, 60% of these species are only in found South Australian waters.

Home to the Marine Bioproducts Cooperative Research Centre, Australia's flagship industry-driven marine bioproducts research and development hub.

Access to world-leading research and development capability.

World-class regulatory framework for aquaculture development.

Marine and land-based licences including macroalgae management areas for seaweed seedstock or broodstock collection to farm seaweed.

Marine areas and coastal land available for farming in a clean, pristine environment.

Established fisheries and aquaculture industries.

Adelaide is one of the world's most liveable cities.

Invest SA - A dedicated team of investment and business professionals to assist your business.

invest.sa.gov.au

SOUTH AUSTRALIA

Government of South Australia

South Australia's rich, unique and diverse marine environment

Australia has the world's third-largest marine area. Much of its rich and diverse marine bioresources are situated within its exclusive economic zone and are yet to be explored and developed for new and improved health, nutrition, aquaculture, agriculture and biomaterial applications.

South Australia is home to a diverse range of marine biota endemic to and only found in South Australia's clean, pristine waters.

The number of seaweed business operating out of South Australia continues to grow - from developing and producing fertilisers, food and health products to traditional aquaculture businesses diversifying into seaweed farming. For example, CH4 Global is working to build a global red seaweed aquaculture business. It has set up its Australian head office in Adelaide and establishing production in South Australia.

The Australian Government has a Seaweed Industry Blueprint outlining the extensive economic, social, and environmental benefits that could be realised through developing an Australian seaweed industry.

Nearly 80% of South Australia's marine biota is unique to our region

Southern Australia has more than 1,400 species of seaweed and up to 60% of these species are found only in South Australian waters.

South Australia has almost 15% of the world's recorded diversity for red and brown seaweeds, which are the most commercially valuable species.

South Australia's world-renowned premium seafood industry

South Australia boasts 3,800km of coastline



South Australia is recognised internationally for its world-class fisheries and aquaculture industries, with a reputation for producing top quality sustainable seafood.

Our coastline covers more than 3,800 kilometres and our clean seas are home to a diverse range of bioresources.

Aquaculture is the fastest growing livestock industry in Australia (9% per annum) to meet increasing global seafood demand. In 2021-22, the South Australian aquaculture industry generated 53% of the State's seafood economic output, worth AUD$238 million.[1] Key commercial aquaculture species include Southern bluefin tuna, yellowtail kingfish, barramundi, oysters, mussels, abalone, microalgae, and seaweed.

Port Lincoln is known as the Seafood Capital of Australia, with one of the largest fishing fleets in the Southern Hemisphere and a major centre for aquaculture development.

[1] *Zoning In: South Australian Aquaculture Report 2023*

South Australia's world-leading research organisations are ready to partner with you to investigate our unique marine biota

Marine Bioproducts Cooperative Research Centre

Australia's largest research and development hub headquartered in Adelaide is dedicated to producing new and sustainable products from the marine environment. The Centre's aim is to drive Australia's transition to the 'third generation' (fishing, aquaculture, bioproducts) of marine production and make our country a global player in the marine biotech industry.

Over the next 10 years, the Marine Bioproducts Cooperative Research Centre (MBCRC) will invest more than AUD$270 million towards marine bioproduct research and development. The not-for-profit centre will bring together 70 partners from academia and industry to work collaboratively on:

- pathways to market for discovered bio-marine products
- creating and expanding sustainable marine bioresource production facilities
- integration of marine science and biotechnology with industries in Australia and overseas
- developing industry-ready graduates.

The MBCRC will focus on production of new sources of marine biomass such as seaweeds, marine micro-algae and filter-feeding animals and the use of advanced manufacturing technologies and processes to produce a suite of novel bioproducts.

South Australian Research and Development Institute

The South Australian Research and Development Institute (SARDI) is the Government of South Australia's principal research institute. SARDI's Aquatic Sciences research division provides technical advice across government, industry and the community.

SARDI offers an integrated research and development capability to sustainably create, nurture and grow aquaculture industries in South Australia. Working closely with the aquaculture sector to produce applied research outcomes, its aquaculture research program is uniquely set up to provide support across the whole spectrum of industry research needs.

With a focus on microalgae and macroalgae, SARDI's algal production sub-program is working on the development of technologies and production systems to underpin and sustain the growth of algal biomass industries in Australia. The program provides clients with concept design, laboratory testing, proof-of-concept and commercial trials. It can also provide practical and cost-effective algal production systems.

The SARDI Aquatic Sciences Environmental and Analytical laboratories provide infrastructure and expertise supporting research into algae strain optimisation, mass cultivation, biomass production and bioprocessing for various end use applications.

The South Australian Aquatic Sciences Centre is a purpose-built marine and freshwater research laboratory complex located in Adelaide. With more than 120 research and technical staff, it is one of the most comprehensive research facilities of its kind in Australia.

Centre for Marine Bioproducts Development, Flinders University

The Centre for Marine Bioproducts Development (CMBD) at Flinders University has been pioneering biorefinery and bioprocessing in marine biotechnology using integrated biotechnology processes to help understand the functional components of raw materials, and transform them into higher value products in a cost-effective manner, whilst reducing waste and pollution. CMBD has collaborated on projects in the areas of biofuels, fish processing co-products, biomaterials, marine drugs, cosmetics and functional foods.

The CMBD has been investigating sustainable processing and commercialisation methods for South Australian microalgae, sea sponges, fish, rock lobsters, sea cucumbers and seaweed. The idea has been to transform these marine bioresources into high-value products for human nutrition, pharmaceuticals and medical materials.

Streamlined approval process for your blue business

South Australia is at the forefront of aquaculture development and planning, and its *Aquaculture Act 2001* is currently the only dedicated aquaculture legislation of its kind in Australia.

South Australia has taken a strategic approach to regulation and seeks to proactively plan for future growth and the expansion of commercial and research activities. While competition for, and access to South Australia's natural resources is increasing, the government is supporting the efficient and effective use of these resources through sound policies and planning and a one-stop-shop approach to aquaculture administration. The objectives of the *Aquaculture Act 2001* are to:

- promote ecologically sustainable development of marine and inland aquaculture
- maximise the benefits to the community from the state's aquaculture resources; and
- ensure the efficient and effective regulation of the aquaculture industry.

The South Australian Government has developed specified Macroalgae Management Areas (MMAs) to ensure seaweed seedstock or broodstock collection is undertaken in a manner where aquatic ecosystems and genetic diversity are maintained.

For further information on the South Australian aquaculture industry, its regulation and zoning policy refer to *Zoning In: South Australian Aquaculture Report 2023*.

South Australia's one–stop–shop approach to aquaculture administration.

To date, the South Australian Government has assessed and approved over 50 aquaculture licence applications to farm various seaweed species across several regions of the state.

Invest SA

Invest SA is a dedicated team of investment and business professionals within the Department of State Development that can connect you with opportunities, industry networks, research and government agencies in the food, wine and agribusiness sector in South Australia.

Our Invest SA, Food, Wine and Agribusiness team offers project facilitation and case management services to develop your business case and to support your business growth.

With our agency partners, Invest SA can support you by:

- offering a dedicated point of contact in state government
- assist with site selection
- ensuring you have the best support to expedite your project; and
- linking and supporting you in discussions with key government decision makers, researchers, government agencies and industry professionals.

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